

Pizza Sales Analysis Documentation

1. Introduction

The **Pizza Sales** project aims to analyze sales data to gain insights into customer preferences, revenue trends, and ordering patterns. This data-driven approach helps optimize inventory, pricing strategies, and marketing efforts for a pizzeria business.

2. Objectives

- Understand **sales performance** by analyzing total revenue and order trends.
- Identify **popular pizza types** and **most common pizza sizes** ordered.
- Determine the **distribution of orders by time** to optimize operations.
- Analyze **category-wise sales distribution** and **revenue contribution** of different pizzas.
- Provide actionable insights for improving business strategies.

3. Data Overview

The analysis is based on SQL queries applied to a structured database containing sales transactions. Queries are categorized into three levels:

Basic Queries

- Retrieve the **total number of orders** placed.

```
select count(order_id)as total_orders from orders;
```

	total_orders
▶	21350

- Calculate the **total revenue** from pizza sales.

```
SELECT
    ROUND(SUM(orders_details.quantity * pizzas.price),
          2) AS total_sales
FROM
    orders_details
    JOIN
    pizzas ON pizzas.pizza_id = orders_details.pizza_id;
```

	total_sales
▶	817860.05

- Identify the **highest-priced pizza**.

```
SELECT
    pizza_types.name, pizzas.price AS highest
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1;
```

	name	highest
▶	The Greek Pizza	35.95

- Find the **most common pizza size** ordered.

```
select quantity, count(order_details_id)
from orders_details group by quantity;
```

```
select pizzas.size, count(orders_details.order_details_id) as order_count
from pizzas join orders_details
on pizzas.pizza_id= orders_details.pizza_id
group by pizzas.size order by order_count desc;
```

	quantity	count(order_details_id)
▶	1	47693
	2	903
	3	21
	4	3

- List the **top 5 most ordered pizza types** and their quantities.

```
SELECT
    pizza_types.name, SUM(orders_details.quantity) AS quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY quantity DESC
LIMIT 5
```

	name	quantity
▶	The Classic Deluxe Pizza	2453
	The Barbecue Chicken Pizza	2432
	The Hawaiian Pizza	2422
	The Pepperoni Pizza	2418
	The Thai Chicken Pizza	2371

Intermediate Queries

- Determine the **total quantity of each pizza category** ordered.

```
SELECT
    pizza_types.category,
    SUM(orders_details.quantity) AS quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY quantity DESC;
```

	category	quantity
▶	Classic	14888
	Supreme	11987
	Veggie	11649
	Chicken	11050

- Analyze the **distribution of orders by hour of the day**.

```
SELECT
    HOUR(order_time) AS hour, COUNT(order_id) AS order_count
FROM
    orders
GROUP BY HOUR(order_time);
```

- | | hour | order_count |
|---|------|-------------|
| ▶ | 11 | 1231 |
| | 12 | 2520 |
| | 13 | 2455 |
| | 14 | 1472 |
| | 15 | 1468 |
| | 16 | 1920 |
| | 17 | 2336 |

- Compute the **category-wise distribution of pizzas**.

```
select category , count(name) from pizza_types
group by category;
```

	category	count(name)
▶	Chicken	6
	Classic	8
	Supreme	9
	Veggie	9

- Calculate the **average number of pizzas ordered per day**.

```
SELECT
    AVG(quantity)
FROM
    (SELECT
        orders.order_date, SUM(quantity) AS quantity
    FROM
        orders
    JOIN orders_details ON orders.order_id = orders_details.order_id
    GROUP BY orders.order_date) AS order_quantity;
```

	AVG(quantity)
▶	138.4749

- Identify the **top 3 most ordered pizza types based on revenue**.

```
select pizza_types.name,
sum(orders_details.quantity * pizzas.price) as revenue
from pizza_types join pizzas
on pizzas.pizza_type_id = pizza_types.pizza_type_id
join orders_details
on orders_details.pizza_id = pizzas.pizza_id
group by pizza_types.name order by revenue desc limit 3;
```

	name	revenue
▶	The Thai Chicken Pizza	43434.25
	The Barbecue Chicken Pizza	42768
	The California Chicken Pizza	41409.5

Advanced Queries

- Compute the **percentage contribution** of each pizza type to total revenue.

```
select pizza_types.category,
round(sum(orders_details.quantity * pizzas.price ) / (select
    ROUND(SUM(orders_details.quantity * pizzas.price),
        2) AS total_sales
FROM
    orders_details
    JOIN
        pizzas ON pizzas.pizza_id = orders_details.pizza_id) *100,2) as revenue
from pizza_types join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join orders_details
on orders_details.pizza_id = pizzas.pizza_id
group by pizza_types.category order by revenue desc ;
```

	category	revenue
►	Classic	26.91
	Supreme	25.46
	Chicken	23.96
	Veggie	23.68

- Analyze the **cumulative revenue trend over time**.

```
select order_date,
sum(revenue) over (order by order_date) as cum_revenue
from
(select orders.order_date,
sum(orders_details.quantity * pizzas.price) as revenue
from orders_details join pizzas
on orders_details.pizza_id = pizzas.pizza_id
join orders
on orders.order_id = orders_details.order_id
group by orders.order_date) as sales;
```

	order_date	cum_revenue
►	2015-01-01	2713.8500000000004
	2015-01-02	5445.75
	2015-01-03	8108.15
	2015-01-04	9863.6
	2015-01-05	11929.55
	2015-01-06	14358.5

- Identify the **top 3 most ordered pizza types per category** based on revenue.

```
select name, revenue from
(select category, name, revenue,
rank() over(partition by category order by revenue desc) as rn
from
(select pizza_types.category, pizza_types.name,
sum((orders_details.quantity)*pizzas.price) as revenue
from pizza_types join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join orders_details
on orders_details.pizza_id = pizzas.pizza_id
group by pizza_types.category, pizza_types.name) as a ) as b
where rn <= 3;
```

	name	revenue
►	The Thai Chicken Pizza	43434.25
	The Barbecue Chicken Pizza	42768
	The California Chicken Pizza	41409.5

4. Key Insights

Sales Performance

- The **total revenue** generated provides an overall measure of business success.
- The **most frequently ordered pizzas** highlight customer preferences.

Customer Preferences

- The **most common pizza size** helps in understanding demand patterns.
- The **top pizza categories** by revenue contribution inform inventory and pricing decisions.

Operational Insights

- Order distribution by hour** helps optimize staffing and preparation times.
- Average daily pizza orders** assist in supply chain and inventory management.

5. Recommendations

1. **Focus on best-selling pizzas:** Increase production of high-demand pizza types.
2. **Optimize pricing strategy:** Adjust prices based on revenue contribution analysis.
3. **Enhance promotions:** Offer discounts on slow-selling pizza categories to boost sales.
4. **Improve operations:** Align staffing with peak ordering hours.
5. **Monitor inventory closely:** Stock more ingredients for popular pizzas to meet demand.

6. Conclusion

The **Pizza Sales Analysis** provides valuable insights into sales performance, customer preferences, and operational efficiency. By leveraging SQL-based data analysis, businesses can make informed decisions to enhance profitability and customer satisfaction.